Accelerating integral decomposition with Kira, machine learning, and exact reductions

Rushil Saraswat

October 19, 2026

Abstract

We study the time required to decompose Feynman integrals using Kira, exact analytic identities, and machine learning. Recognized vacuum factors are evaluated by partial fractions and analytic tadpole formulas, while connected one-mass trees return explicit Gamma-function values. For supported remaining two-loop boxes, we train a measured-cost selector for sparse and complete seed regions, including failed attempts in its objective. Validation rejects the new complete-region selector, so the final application retains the preceding native seed model. A frozen comparison tests 69 requests, with vacuum products and connected trees at every loop order from one through thirteen. The integrated and preceding applications complete 69 and 69 requests. Their 69 matched requests take 109.273 and 111.950 seconds, respectively. On 16 matched native batches, the integrated-to-fixed time ratio is 1.002, with a 95% interval of $[0.994, 1.012]$. A separate analytic coverage grid completes 26 of 26 requests. We report analytic and learned effects separately and retain native row selection because earlier learned row policies did not establish a reliable benefit. The results apply to the declared domains and do not establish a speedup for every high-loop scattering integral.

1 Introduction

Feynman integrals arise from integration over internal loop momenta in higher-order perturbative corrections to scattering amplitudes and particle decay amplitudes. Many integrals in a calculation share propagators and differ only in their powers or numerators. Integration-by-parts (IBP) identities relate these integrals and express them in terms of a smaller set of master integrals [1, 2]. This decomposition supplies the coefficients needed to assemble an amplitude; the evaluation of any remaining master integrals is a subsequent calculation [1].

Kira automates the generation and solution of the IBP system, and its current methods already remove unnecessary seeds and equations [3, 4]. The remaining cost depends on the integral structure and the requested targets. Generating fewer useful equations can reduce work, but a restricted system can also fail to reach the required basis and require a larger calculation. We therefore optimize complete reduction time, including prediction, algebra, and recovery. Seed-count reduction alone does not measure this objective.

Our method first identifies structures that admit an exact reduction formula. It applies those formulas to complete requests, individual targets, and recognized intermediate sectors. For the remaining supported native calculations, we compare trained cost selection with a fixed seed region before Kira generates the identities. Exact algebra determines every output coefficient. Related machine-assisted searches motivate the seed intervention [5, 6], while analytic one-loop reduction motivates the formula stage [7, 8]. We derive the particular formulas and cost objective used here in the appendices.

The present extension evaluates vacuum factors with repeated masses and raised or negative denominator powers by exact partial fractions. It also returns explicit Gamma-function values for the previously reduced connected one-mass trees. For native work, we train a new cost selector for

sparse and complete seed regions, including the time lost to unsuccessful attempts. We compare the integrated method with the preceding application and with a fixed choice. These controls separate the contribution of analytic formulas from the contribution of ML. Earlier equation-selection and row-selection results remain separate, and the companion notes preserve their full experiments.

2 The decomposition problem and Kira

We first define the notation used in this paper. An integral family specifies inverse propagators, masses, and external scalar products. In dimensional regularization, we write [1, 3]

$$I_{\boldsymbol{a}}(\boldsymbol{x}, d) = \int \prod_{h=1}^{L} \frac{d^d \ell_h}{i\pi^{d/2}} \prod_{j=1}^{N} D_j^{-a_j}, \qquad D_j = Q_j^2 - m_j^2 + i0, \qquad d = 4 - 2\epsilon. \tag{1}$$

Here L is the number of loops, ℓ_h is loop momentum h, N is the number of integral coordinates, and Q_j is a specified linear combination of loop and external momenta. The quantity m_j is the mass on line j, $\boldsymbol{x}$ contains squared masses and external invariants, ϵ is the dimensional regulator, and $i0$ specifies the Feynman pole prescription. A positive integer a_j gives a denominator power, zero removes that factor, and a negative integer gives a numerator power. Auxiliary inverse propagators complete the coordinates for loop scalar products when the physical lines do not suffice [3].

A sector consists of integrals with the same positive-index positions. Removing a positive position gives a sub-sector. A seed $\boldsymbol{\sigma}$ is an index vector at which we generate identities [3]. For a loop or external momentum v, the IBP relation is [2]

$$0 = \int \prod_{h} \frac{d^d \ell_h}{i\pi^{d/2}} \frac{\partial}{\partial \ell_k^\mu} \left(v^\mu \prod_j D_j^{-\sigma_j} \right) = \sum_u A_{\boldsymbol{\sigma},o;u}(\boldsymbol{x}, d) I_{\boldsymbol{a}^{(u)}}. \tag{2}$$

The operator label o specifies (k, v), and $A_{\boldsymbol{\sigma},o;u}$ is the coefficient of integral u. Appendix A expands the derivative into index shifts and explains why the coefficients are rational. Each seed supplies several operators and therefore several equations.

Kira orders integrals, uses symmetries and other valid relations, selects equations, and performs elimination and back-substitution [3, 4, 9]. For a target T, the result has the form

$$T = \sum_{m=1}^{n_M} r_{Tm}(\boldsymbol{x}, d) M_m, \qquad r_{Tm} \in \mathbb{Q}(\boldsymbol{x}, d). \tag{3}$$

Here M_m is master integral m, n_M is the number of masters, and $\mathbb{Q}(\boldsymbol{x}, d)$ is the field of rational functions with rational coefficients. Exact row operations give Eq. (3), as derived in Appendix A. A remaining free integral in a restricted system does not by itself establish a master of the full family. We check completion against a reference basis. Kira can solve symbolically or use finite fields and reconstruct rational functions [10]; this study leaves its reconstruction algorithm unchanged.

The three learned interventions occur at distinct points (Table 1). A seed policy changes which identities are generated. An equation policy selects seed–operator pairs. A row policy chooses among allowed elimination steps after coefficients exist. All three can change cost without predicting the physical coefficients themselves.

3 Exact reductions before native elimination

The exact stage uses the propagator structure to determine whether a formula applies. For independent loop components, the product measure and separated integrand factorize Eq. (1);

Table 1: Learned decisions studied here. The last column states their use in the measured method.

Decision	Work affected	Use in this study
Seed region	Generation and solution of the IBP system	Retained seed-cost model; complete-region replacement rejected by validation
Equation retention	Construction of selected IBP coefficients	Separately validated native filter
Elimination row	Intermediate expressions and row operations	Omitted from the retained application

Appendix A proves the resulting product of component reductions. Identical components share their calculation. Connected graphs require their own identities, so a high loop count alone does not select this shortcut.

Table 2 summarizes the retained domains. For generic one-loop sectors, we resolve numerator scalar products along the external directions and average the perpendicular components. This yields scalar integrals whose raised powers are lowered by an exact IBP recurrence. Appendix F derives both steps. Their coefficients are rational in d, and keeping known linear dimension factors separate reduces repeated symbolic simplification. A singular matrix can prevent those particular recurrences from being used, which motivates the new two-point rule.

Table 2: Exact identities used by the application. Bounds describe the implementation and do not classify all reducible integrals.

Integral structure	Identity and accepted domain
Generic one-loop sectors	Tensor angular average and scalar IBP recurrence; rational kinematics, at most five coordinates, numerator and excess positive powers at most six; required matrices invertible
Zero-invariant two-point sectors	New endpoint formula; equal, unequal, or zero masses; rational kinematics, at most five coordinates, numerator and excess positive powers at most four
Scaleless vacuum families	Homogeneity gives zero; no masses or external momenta, up to 128 coordinates and thirteen loops
Vacuum factors	Partial fractions and analytic tadpoles; rational nonnegative masses, up to eight denominators per factor, and products through thirteen loops
Symbolic two-point masses	New scalar mass-derivative formula at $p^2 = 0$; individual powers at most six and total power at most eight
Connected one-scale trees	Successive massless bubble integration; massless two-point root or one massive vacuum root; implemented through thirteen loops
Independent products	Product of accepted component reductions; repeated components computed once

3.1 Two-point sectors at zero external invariant

Let $D_1 = \ell^2 - x$ and $D_2 = (\ell - p)^2 - y$, where x and y are squared masses and $p^2 = 0$. Write $T(x)$ for the unit-power tadpole with squared mass x. Feynman parametrization gives

$$B_{11}(x, y) = \frac{T(x) - T(y)}{x - y}, \qquad B_{ab}(x, y) = \frac{\partial_x^{a-1} \partial_y^{b-1} B_{11}(x, y)}{(a-1)!(b-1)!}. \tag{4}$$

Here a and b are positive propagator powers, and derivatives treat the two masses as independent before any specialization. Appendix G derives this relation, its equal-mass limit, and the finite tensor formula. At equal masses, $B_{ab}(x, x) = T_{a+b}(x)$, where T_n is the tadpole with denominator power n. A zero-mass tadpole vanishes in dimensional regularization [1].

For polynomial numerators, the same parameter shift leaves one radial denominator. Angular moments reduce the numerator to a finite sum, and a change from the Feynman parameter to the interpolated squared mass expresses each term through the two endpoint tadpoles. The result is a rational coefficient of $T(x)$ and $T(y)$, with no inverse external Gram matrix. This also applies when a generic one-loop recurrence reaches such a two-line sub-sector. Applying the identity at that intermediate step extends the complete reduction beyond requests that begin as bubbles.

The application tests the stated conditions before using a formula. It retains native reduction for unsupported targets, symbolic numerators outside the accepted domain, and recurrence limits. A request can contain both accepted and remaining targets; the application combines their exact outputs. These checks determine mathematical applicability. They do not ask a learned model to certify an identity.

3.2 Direct vacuum evaluation

A massless vacuum family has no dimensional scale and vanishes in dimensional regularization [1]. We apply this check before any elimination, including to coupled loop momenta. Appendix I gives the radial scaling argument and the implemented conditions.

When every denominator in a component carries the same loop momentum, its integrand is a rational function of that momentum squared. Equal masses combine their powers, and partial fractions express the remaining poles through raised-power tadpoles. Appendix I derives the coefficients and evaluates each tadpole through Gamma functions. Polynomial terms and massless vacuum factors vanish in dimensional regularization [1]. The application therefore returns both a master decomposition and an explicit analytic value for this domain.

This rule also handles repeated masses and polynomial numerators. It applies independently to each separated loop factor and computes identical components once. Recognition requires the absence of a momentum coupling between factors. Connected graphs use a separate identity.

3.3 Connected trees and higher loop counts

Successive massless bubble integrations also reduce specific connected trees to a unit-power integral of the same tree. For a massless two-point root, the coefficient is a rational function of d and the external invariant. For a massive vacuum root, it is rational in d and the one squared mass. Appendix H derives these statements from Gamma-function products and their integer shifts. Equation (68) gives the complete analytic value of a connected vacuum tree, and the application now returns this value with its reduction. This family includes connected high-loop integrals, while independent products provide a separate high-loop test. Neither construction represents general coupled multiscale scattering.

4 Learning the remaining Kira cost

We test a replacement for the preceding native seed model. Its candidate procedures are: a tight sparse region, a wider sparse region, or every seed in the finite bounds used by the tight procedure. Appendix B defines the sparse inequalities. The complete-region action removes that sparse restriction while retaining the same Kira algebra and basis checks. It can avoid the cost of an unsuccessful restricted attempt. Recovery remains available if any chosen region is insufficient.

The retained native model uses the wide, tight, and medium sparse regions of Appendix B. It was trained on 34 batches and selected with sixteen validation batches. Its single cost tree has maximum depth two, and its failure guard uses twelve depth-three trees per action. These weights remain unchanged. The new complete-region replacement described here is retained only if validation supports it.

The replacement training set contains 24 planar and nonplanar box batches, and a separate set of twelve batches selects the model. Each batch contains two to six targets with numerator rank one through four. Twenty-one features describe target counts, numerator distributions, and physical propagator incidence. They contain neither family names nor numerical kinematic profiles. Whole batches are disjoint across development, training, validation, and testing, although individual target vectors can recur. Appendix J specifies the four sets.

We measure each replacement training and validation action twice and train on its median complete time. This includes unsuccessful preliminary reductions and recovery. Cost-sensitive trees and bootstrap ensembles follow the recursive constructions of Refs. [11, 12]; Appendix C derives the cost objective and lists the candidate settings. Validation selects a fixed control and enables a learned choice only when it improves the measured cost by at least one percent with no unverified selected results. An action that fails its exact checks receives a training penalty of at least 60 seconds.

A separate classifier estimates action failure from the training outcomes. Inputs outside the observed feature range or actions above the selected risk threshold use the fixed control, which must complete every training and validation request. A predicted score is an empirical decision aid, so the application still checks completion. This design trains against measured full cost while leaving every integral coefficient to exact algebra. A complete seed region can avoid an insufficient restricted system, but it can also increase the algebraic work. The timing objective accounts for both effects.

4.1 Equation and row policies

The equation-selection model ranks candidate seed–operator pairs before constructing their full coefficients. Its independently tested filter reduces complete time on its recorded workload. The current combined comparison keeps that filter off, so its benefit is reported separately. Multiplying the seed and equation speed ratios would assume an interaction that these measurements do not test.

The original next-row model scores allowed elimination steps in Kira's changing system. Its inference and selection costs exceeded the algebraic savings in the retained comparison. A later compiled cost policy reduced this overhead, but its interval still included equal time. We therefore omit learned row selection from the final application and retain Kira's exact elimination rules. Section 6 reports the corresponding measurements.

5 Comparison design

We freeze the formulas, learned policy, fixed action, and reserved requests before testing. After validation rejects the replacement, we freeze the application that retains the preceding native model. The final comparison contains 69 fresh target batches: thirteen two-mass vacuum products, 34 connected one-mass vacuum trees, three zero-invariant tensor bubbles, three ordinary one-loop tensors, and sixteen native box batches. Both vacuum grids cover every loop order from one through thirteen. The connected grid includes chains, stars, and binary trees, with coincident small-loop routings omitted. Separate tests of the unchanged exact routines contain 26 requests: repeated-mass factors and coupled scaleless vacuum families, with one request per domain at each of the same loop orders. Their coverage records are retained from the preceding candidate

experiment.

An independent symbolic partial-fraction calculation supplies the vacuum-product references. Exact rational identities, convergent radial integrals, and a native tadpole recurrence check provide additional tests. Native box, zero-invariant, and ordinary tensor requests use independently prepared Kira references. The connected-tree checks compare with the preceding exact implementation and preserve the independently derived identity. Appendix J states these distinctions.

The compared methods are the integrated application, the same application with its selected fixed native action, and the preceding published application. We use the same input and output procedure for all methods, and the preceding control calls the unchanged preceding reduction routine. Every timed run starts a fresh process and includes imports, model loading, prediction, exact formulas, native equation generation and solution where used, recovery, output, and exit. We use one computational thread, three repetitions, and rotating method order on the same worker. Reference preparation and the final external coefficient comparison lie outside the timer. Native applications receive protected master indices where required; reference coefficients are used only for the external check. These times therefore assume the stated basis.

Each application has a 30-second limit. A request counts as complete only when every repetition agrees with its reference. The time ratio uses the sum of per-request median times on the matched completed subset. All incomplete requests remain in the completion counts. Appendix E defines this estimator and the request-resampling interval. The separate repeated-mass grid reports completion and time without claiming a speed factor against an unsupported control.

6 Measured results

The following values are measurements from the frozen comparison in this study. Table 3 separates the physical domains and includes completion counts. Summed median times refer only to the matched completed requests.

Table 3: Complete times against the preceding application. Done gives preceding/integrated completions. Each domain uses the same requests for both methods.

Domain	Done	Paired	Preceding (s)	Integrated (s)
Two-mass products, 1–13 loops	13/13	13	2.271	1.935
Connected vacuum trees, 1–13 loops	34/34	34	7.900	5.719
Zero-invariant tensor bubbles	3/3	3	0.577	0.569
Ordinary one-loop tensors	3/3	3	0.688	0.698
Native two-loop boxes	16/16	16	100.514	100.353
All 69 requests	69/69	69	111.950	109.273

On the 69 matched requests, the integrated-to-preceding time ratio is 0.976, with a central 95% request-resampling interval of $[0.952, 0.990]$. The point estimate corresponds to 2.4% less time. The integrated application is slower on 15 matched requests. This is an incremental comparison with the preceding application on these inputs, rather than a new comparison with unmodified Kira. Earlier percentages use other workloads and cannot be added to this change.

6.1 Direct formulas and their high-loop scope

The two-mass product grid changes from 2.271 to 1.935 seconds on 13 matched requests. Direct partial fractions avoid generic one-loop reduction and its preparation for these inputs. The connected vacuum grid changes from 7.900 to 5.719 seconds on 34 matched requests. The connected identity is preserved, while its early dispatch avoids generic symbolic preparation when

the simpler tree formula already applies. Both grids contain every loop order from one through thirteen.

The separate analytic coverage grid completes 26 of 26 requests and takes 5.983 seconds in summed median time. Its thirteen repeated-mass requests use an independent symbolic partial-fraction reference, while thirteen coupled massless vacuum requests use the scaleless-integral identity. We report this as exact-domain coverage without assigning a speedup over an unsupported native input. The implementation returns Gamma-function values for the resulting tadpoles and the connected one-mass tree masters, so these domains have an analytic value as well as a decomposition.

The tests contain 14,922 coefficient comparisons across timed outputs. Development additionally checks 48 rational-function identities, 138 product coefficients, 4 native tadpole coefficients, 6 native scaleless target relations, and two convergent radial integrals. These checks test algebra and normalization separately.

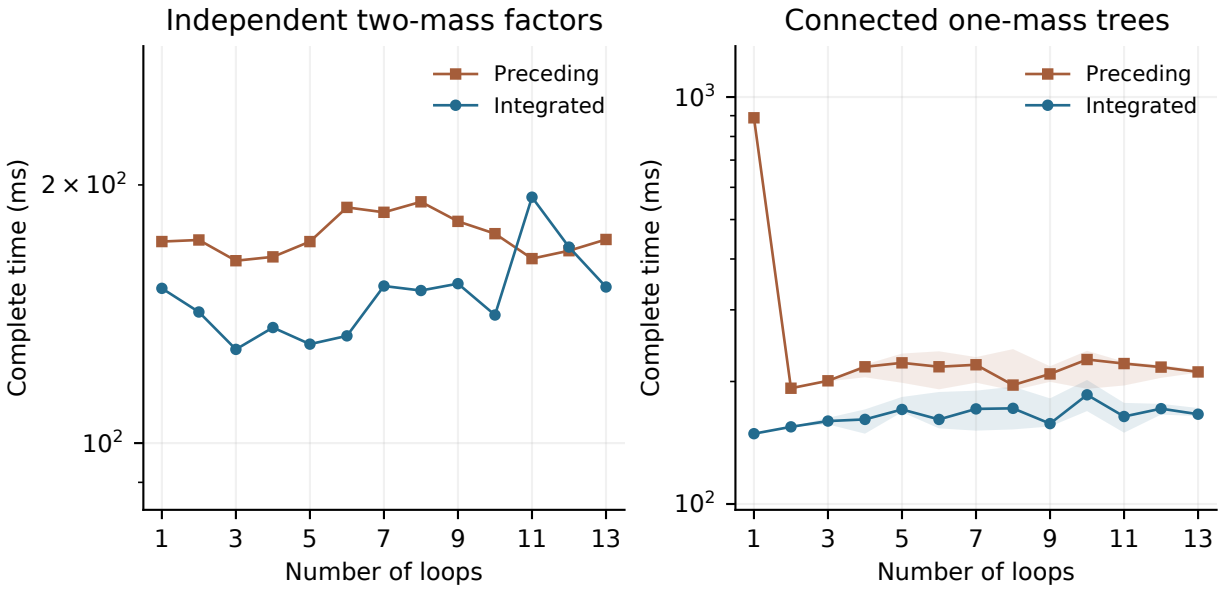


Figure 1: Direct formulas reduce aggregate time in both measured vacuum grids. The panels show complete elapsed time at every tested loop order, with independent factors on the left. In the connected-tree panel, each line gives the median across the tested shapes at that loop order and the shaded band gives their full range. Logarithmic time axes retain the short exact calculations and the slower preparation costs. These domains require the structures stated in the derivations.

6.2 Native seed choices and the ML contribution

The preliminary ordering experiment verifies all 32 development reductions. Ordering 1 has the lowest sum of development times, so the final native actions retain it. The subsequent model chooses the seed procedure. Validation selects 1 cost tree with maximum depth 1 and a minimum leaf population of 2. Its fixed control is the tight sparse region. The selected validation cost is 86.114 seconds, including the prediction allowance and failure penalties; the fixed control gives 86.090 seconds. Validation rejects the replacement because the deployment condition is not met. The final application retains the preceding validated native seed model.

On the 16 matched reserved native batches, the fixed application takes 100.166 seconds and the integrated application takes 100.353 seconds. Their ratio is 1.002, with interval $[0.994, 1.012]$. This interval includes equal time and does not resolve a consistent additional learned saving. The integrated method is slower on 7 matched batches. The retained native policy changes the fixed action on 0 of the sixteen requests in the final repetition. Thus the retained model selects the

fixed action throughout this final sample; the measured difference includes prediction overhead and run-to-run variation, and supplies no additional learned speedup. Median measured model loading and decision time is 2.661 milliseconds. Exact formulas, the native interface, and recovery checks are shared by the learned and fixed controls.

Across 48 native applications per method, unverified output counts are 0 for the integrated application, 0 for the fixed control, and 0 for the preceding application. Recorded failed inner attempts total 0, 0, and 0, respectively. A terminated application can lack an inner-attempt record and remains in the separate application-failure count. Complete-region selection can avoid a sparse failure, but these measurements do not guarantee success for unseen inputs.

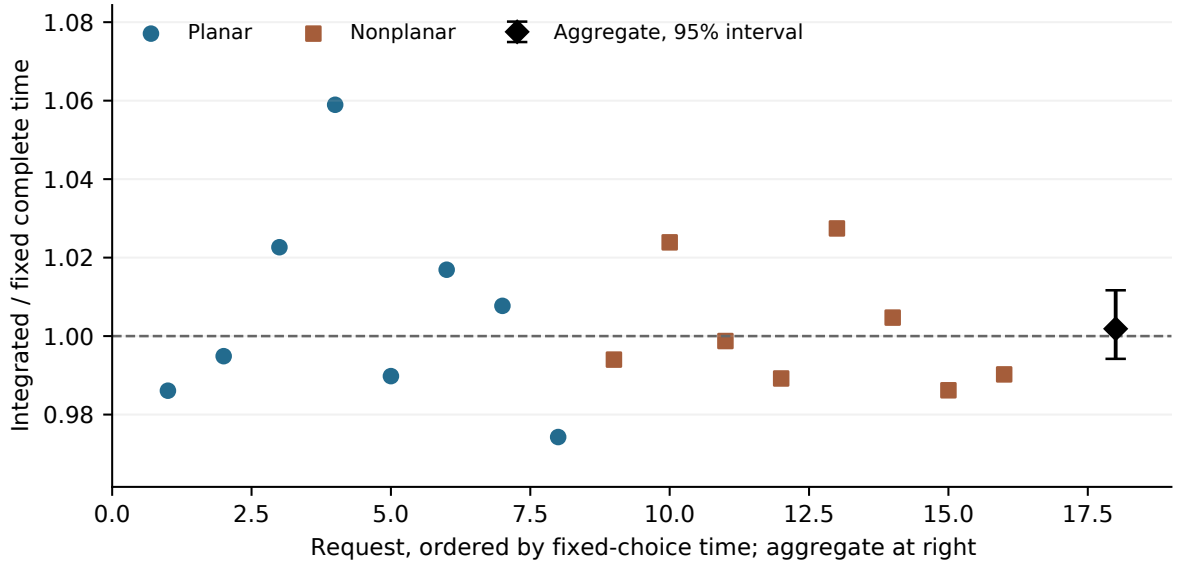


Figure 2: The retained model chooses the fixed action on every final batch, so this sample demonstrates no additional ML saving. Both controls use the same exact methods and native implementation. Points below one indicate a lower measured integrated time. Individual batches show variation, while the aggregate point reports the ratio of summed median times and its request-resampling interval.

6.3 The other learned interventions

Table 4 retains the main earlier controls needed to interpret the method. They use different workloads and cannot be multiplied into a cumulative speed factor. The equation-filter result supports that intervention on its tested domain, while the row results motivate its omission. The current combined experiment keeps equation filtering off and does not claim a joint seed-plus-equation gain. The earlier seed comparison has interval $[0.937, 0.984]$, and the equation

Table 4: Earlier measurements from this study, retained as separate evidence. A ratio below one means less time. Their full input definitions and measurements remain in the notes.

Intervention	Control and workload	Time ratio
Earlier native seed model	Fixed region, 32 box batches: 156.383 to 150.663 s	0.963
Equation selection	Native filter, 24 new-index batches: 72.295 to 67.478 s	0.933
Original learned row model	Native row choices, 84 trials	1.305
Compiled row-cost policy	Native row choices; geometric mean	0.982

comparison has interval $[0.928, 0.940]$. The compiled row policy has interval $[0.959, 1.003]$, which

includes equal time. These intervals use the resampling procedure recorded for each original study. The native row algorithm remains in the retained workflow.

6.4 Scope of the result

The direct formulas prove reductions and analytic values within their stated domains. The test covers every loop order through thirteen for independent vacuum products and connected one-mass trees. The native test covers new target batches and kinematic points within two known two-loop scattering graphs. General coupled multiscale scattering at every loop order remains outside this evidence. Loop count alone does not classify the difficulty or determine whether a particular exact rule applies.

The controls separate changes in the integrated application from the additional effect of learned selection. A finite search over models and native settings cannot establish globally minimum time for every integral. The retained measurements include slower requests, unsuccessful attempts, and incomplete applications so that these limits remain part of the result.

7 Conclusion

Exact identities reduce complete calculation time when the propagator structure admits them. The added partial-fraction rule handles repeated masses, raised powers, and polynomial numerators in vacuum factors. Its tadpole formula and the retained connected-tree derivation also supply explicit analytic values. Direct dispatch avoids algebraic preparation that these identities make unnecessary.

The integrated method completes 69 of 69 reserved requests. Its 69 matched requests take 109.273 seconds, compared with 111.950 seconds for the preceding application. The separate native comparison with a fixed seed procedure gives ratio 1.002 and interval $[0.994, 1.012]$. These are distinct controls: the first measures the full revision, and the second tests the additional learned choice. Validation rejects the new complete-region selector, so the final application retains the preceding native seed model. Earlier equation filtering remains separately supported on its tested workload, while learned row selection is omitted.

The high-loop evidence covers products and connected one-scale trees through thirteen loops. Native generalization is tested within planar and nonplanar two-loop boxes. General coupled multiscale high-loop scattering and globally minimum reduction time remain unestablished. The notes preserve the earlier experiments, unsuccessful choices, and full coverage limits.

A IBP identities, row operations, and exactness

The starting total-derivative identity follows from integration by parts in a region where the integral converges, followed by dimensional analytic continuation [2, 1]. Write $P_{\boldsymbol{\sigma}} = \prod_i D_i^{-\sigma_i}$ and $\partial_k = \partial/\partial\ell_k$. The product rule gives

$$\partial_k \cdot (v P_{\boldsymbol{\sigma}}) = (\partial_k \cdot v) P_{\boldsymbol{\sigma}} - \sum_i \sigma_i (v \cdot \partial_k D_i) D_i^{-\sigma_i - 1} \prod_{j \neq i} D_j^{-\sigma_j}. \tag{5}$$

For $v = \ell_h$, the first factor is $d\delta_{kh}$; for an external momentum it is zero. Completeness of the inverse-propagator coordinates permits us to write $v \cdot \partial_k D_i = u_i + \sum_j v_{ij} D_j$, where u_i and v_{ij} are coefficients fixed by the propagators and kinematics, with the invariant and mass terms included in u_i. Integrating Eq. (5) therefore gives

$$0 = (\partial_k \cdot v) I_{\boldsymbol{\sigma}} - \sum_i \sigma_i u_i I_{\boldsymbol{\sigma} + \boldsymbol{e}_i} - \sum_{i,j} \sigma_i v_{ij} I_{\boldsymbol{\sigma} + \boldsymbol{e}_i - \boldsymbol{e}_j}. \tag{6}$$

The unit vector $\boldsymbol{e}_i$ raises index i by one. This derivation explains how a seed and an operator generate a row of the reduction system.

Let q be a source row and k a pivot column with $A_{qk} \neq 0$. Solving the identity $\sum_j A_{qj} I_j = 0$ for I_k gives $I_k = -\sum_{j \neq k} (A_{qj}/A_{qk}) I_j$. Subtracting A_{rk}/A_{qk} times that zero identity from row r yields

$$A'_{rj} = A_{rj} - \frac{A_{rk}}{A_{qk}} A_{qj}, \qquad A'_{rk} = 0. \tag{7}$$

Every step preserves the represented relation. Repetition until only prescribed master columns remain gives Eq. (3). Removing seeds or equations preserves the validity of the retained identities, but can prevent this elimination from finishing. We therefore require reduction to the prescribed basis and exact agreement with the reference coefficients. Generic kinematics avoid singular pivot denominators; special kinematic limits require their own treatment.

For independent components, write the loop momenta as disjoint groups and the integrand as $\prod_b f_b$. In a convergence domain, repeated integration gives $I = \prod_b \int f_b$. Dimensional analytic continuation preserves that identity [1]. Substituting each exact component reduction therefore gives

$$I = \prod_b \left(\sum_m r_{bm} M_{bm} \right). \tag{8}$$

Our product output retains this factored form. It avoids expanding the Cartesian product of master terms and uses the same representation in all timed methods.

B The seed regions and equation score

For seed $\boldsymbol{\sigma}$, define excess positive power $D = \sum_i \max(\sigma_i - 1, 0)$, numerator rank $R = \sum_i \max(-\sigma_i, 0)$, and positive sector $P = \{i : \sigma_i > 0\}$. Let h be the number of absent physical lines relative to the largest top sector, and let D_T, R_T be the largest target values. The distance from the target numerator rectangles is defined by

$$q(\boldsymbol{\sigma}, \mathcal{T}) = \min_{\boldsymbol{a} \in \mathcal{T}} \sum_{i \notin P} \max\{0, -\sigma_i - \max(0, -a_i)\}. \tag{9}$$

Projection onto the interval from zero to each target numerator power leaves precisely this positive excess. Summing the distances and minimizing over targets gives Eq. (9). We define the region by

$$\begin{aligned} q &\leq \max\{0, \lfloor \theta_0 + \theta_1 h + \theta_2 D + \theta_6 R \rfloor\}, \\ R &\leq \max\{0, \lfloor R_T + \theta_3 - \theta_4 h \rfloor\}, \qquad D \leq \max\{0, \lfloor D_T + \theta_5 \rfloor\}. \end{aligned} \tag{10}$$

These inequalities specify a search family rather than a sufficiency theorem. Native outer bounds, retained sector corners, and admissible target and master seeds also apply. The previously studied sparse choices use

$$\begin{aligned} \boldsymbol{\theta}_{\text{wide}} &= (2.5, 0, 0, 1.25, 0, 1, -0.25), \\ \boldsymbol{\theta}_{\text{tight}} &= (1, 0, 0, 0, 1, 1, 0), \\ \boldsymbol{\theta}_{\text{medium}} &= (1.5, 0, 0, 0.5, 0, 1, 0). \end{aligned} \tag{11}$$

The retained native model uses these three sparse choices. The rejected replacement uses the wide and tight choices and a third action that retains all seeds within the tight procedure's native outer bounds. The native procedure retains its subsector truncation and recovery rules. The model selects one complete procedure before generating equations.

The separately tested equation classifier has seventeen features: four family and request dimensions; three seed counts; two differences from target power maxima; three sector descriptors; exact target membership and minimum index distance; the operator index; and the two target maxima. Positive labels mark equations retained by a complete reference reduction. Sampling at most 2,000 records of each class y in batch c gives each sampled record weight $N_{cy}/(n_{cy}N_c)$, where N_{cy} and n_{cy} are the native and sampled class counts and N_c is the full batch count. Multiplying this weight by n_{cy} preserves the class fraction N_{cy}/N_c. A leaf with weighted totals W_0, W_1 returns $p = W_1/(W_0 + W_1)$, and a fixed threshold retains equations. The decision-tree construction follows Ref. [11]. This empirical score does not certify sufficiency or probability calibration; the native completion check remains necessary.

C Measured-cost learning

Let C_{ia} be the measured complete time of training request i under action a. Unverified actions receive $C_{ia} \geq 60$ seconds. This is a training penalty, and test failures are reported as failures. A leaf containing requests S assigns action cost

$$\widehat{C}_a(S) = \frac{1}{|S|} \sum_{i \in S} C_{ia}, \qquad a_S = \arg\min_a \sum_{i \in S} C_{ia}. \tag{12}$$

The second formula follows because the first denominator does not depend on the action. A proposed split $S = S_1 \cup S_2$ improves the empirical chosen cost by

$$\Delta(S_1, S_2) = \min_a \sum_{i \in S} C_{ia} - \min_a \sum_{i \in S_1} C_{ia} - \min_a \sum_{i \in S_2} C_{ia}. \tag{13}$$

This is the parent cost minus the sum of child costs. We require $\Delta > 0.002|S|$ seconds before splitting. This cost-sensitive objective uses the recursive tree and bootstrap-ensemble constructions of Refs. [11, 12].

For each target, let u and v be its two nonnegative auxiliary numerator powers and let $R = u+v$. The 21 inputs are batch size; minimum, maximum, and mean R; number of distinct R values; minimum, maximum, and mean of each of u, v; counts with $u = 0$ and $v = 0$; sums of uv and $(u - v)^2$; number of physical propagators containing both loops; largest number of physical propagators incident on a loop; and counts with $R = 1, 2, 3, 4$. All are available before reduction.

We compare 1, 12, or 32 trees, maximum depths 1 through 5, minimum leaf populations 2, 4, or 8, and required fractional savings $\eta = 0, 0.03, 0.08, 0.15$. An ensemble averages leaf cost vectors. A single tree uses the complete training set; each ensemble tree samples training requests with replacement. The fixed action b minimizes total validation cost among actions that completed every training and validation request. If a_* has the least predicted cost, we choose it only when

$$\widehat{C}_{a_*} + 0.002 < (1 - \eta)\widehat{C}_b. \tag{14}$$

The allowance is two milliseconds per prediction during selection. Equation (14) is the defined minimum predicted saving, obtained by adding prediction overhead to the selected cost. Validation compares candidates by unverified count, complete cost, tree count, and depth, in that order. Deployment additionally requires at least one percent less validation cost and a change from the fixed action. We also fit a failure classifier for each action from its verified/unverified training labels. Each classifier has twelve trees of depth at most three and at least two requests per child. One tree uses all requests, and eleven use bootstrap samples. A leaf returns its failed fraction p_f. Binary Gini impurity for n requests with k failures is

$$n\left[1 - (k/n)^2 - (1 - k/n)^2\right] = \frac{2k(n-k)}{n}. \tag{15}$$

The equality follows by expansion. Splits maximize the decrease in this quantity, following the classification-tree construction of Ref. [11]. Averaging the leaf fractions gives an empirical risk score for each action, without a claim of calibrated failure probability. Validation selects a threshold from $0, 0.05, 0.15, 0.30$ jointly with the cost policy. If the proposed score exceeds that threshold, or any feature lies outside its training range, the policy selects the fully verified fixed region. Validation requires no unverified selected actions before enabling the model. This guard uses observed evidence; a zero score is not a proof that an unseen reduction will complete. The application timer includes both predictions and any subsequent recovery.

D Physical test inputs

All profiles below define squared quantities in a common fixed unit. The zero-invariant families use shifts $(0, p, q, r)$ and masses $(x, y, 0, 0)$, with

$$\begin{aligned} &p^2 = 0, \quad p \cdot q = -h - 1, \quad p \cdot r = -h - 2, \\ &q^2 = -2h - 7, \quad q \cdot r = -h - 3, \quad r^2 = -3h - 13. \end{aligned} \tag{16}$$

The unequal case has $(x, y) = (h+3, h+11)$, the equal case $(h+3, h+3)$, the mixed case $(h+3, 0)$, and the massless case $(0, 0)$. Targets have the form $(1+\delta_1, 1+\delta_2, -b_1, -b_2)$, where $\delta_1 + \delta_2$ is the excess positive power and $b_1 + b_2$ is numerator rank.

Ordinary tensor families have P physical lines, one auxiliary coordinate, shifts $q_1 = 0$, $q_{i+1} = p_i$, and

$$m_i^2 = h+11i^2 \ (i \le P), \quad m_{P+1}^2 = 0, \quad p_i^2 = -(10h+17i), \quad p_i \cdot p_j = -(h+i+2j-2) \ (i < j). \tag{17}$$

Appendix J specifies the current profiles, targets, and separate vacuum grids.

For the native planar box, let p_1, p_2, p_3 be massless, $2p_1 \cdot p_2 = s$, $2p_1 \cdot p_3 = t$, and $2p_2 \cdot p_3 = -s-t$. The seven physical line momenta are

$$(\ell_1, \ell_1 - p_1, \ell_1 - p_1 - p_2, \ell_2, \ell_2 - p_1 - p_2, \ell_2 - p_1 - p_2 - p_3, \ell_1 - \ell_2), \tag{18}$$

and the auxiliary momenta are $(\ell_1 - p_3, \ell_2 - p_1)$. The nonplanar routing follows Ref. [3]:

$$\begin{aligned} (Q_1, \ldots, Q_9) = (&\ell_1, \ell_2, p_1 - \ell_1, p_2 - \ell_2, p_1 + p_2 - p_3 - \ell_1, \\ &p_1 - \ell_1 + \ell_2, p_1 - p_3 - \ell_1 + \ell_2, \ell_1 - p_3, \ell_2 - p_1 - p_2). \end{aligned} \tag{19}$$

The first seven entries are physical and all masses vanish. Its convention gives $p_i^2 = (p_1+p_2-p_3)^2 = 0$, $s = (p_1+p_2)^2$, and $t = (p_1-p_3)^2$. Expanding the squares gives $2p_1 \cdot p_3 = -t$ and $2p_2 \cdot p_3 = s+t$. The planar profiles use $(s, t) = (-1009 - 12h, -1021 - 14h)$ and the nonplanar profiles use $(-113 - 18h, -127 - 10h)$.

Every native target is $(1, 1, 1, 1, 1, 1, 1, -u, -v)$ with $u, v \ge 0$ and $1 \le u + v \le 4$. The current split uses the profiles and batch counts in Appendix J. Both graph types occur in all splits, so the test measures new batches and kinematic points within these structures.

E Time ratios and uncertainty

Let T_{imr} be complete elapsed time for request i, method m, and repetition r. For the set $\mathcal{C}$ completed by both methods in all repetitions, define

$$S_m = \sum_{i \in \mathcal{C}} \text{median}_{r=1,2,3}\, T_{imr}, \qquad R = \frac{S_{\text{new}}}{S_{\text{control}}}, \qquad \text{fractional saving} = 1 - R. \tag{20}$$

These definitions compare identical completed requests. Dividing both times by the control time gives R, and subtracting gives the fractional saving. Completion counts remain separate because an unfinished reduction has no verified finite speedup. To estimate variation across requests, we draw $|\mathcal{C}|$ paired request indices with replacement and recompute R. We report the central 95% of 4,000 such ratios, using the bootstrap construction of Ref. [13]. This interval describes the sampled request collection and does not establish performance for every integral family.

F Exact reduction of polynomial numerators

F.1 Resolving the loop momentum

One-loop tensor reduction expresses loop-momentum numerators through scalar integrals. Established methods include tensor decompositions and generating functions for arbitrary numerator and denominator powers [7, 8]. We derive the particular identities used in our extension here. The algebra uses rational kinematic values and keeps d symbolic. It applies to a one-loop sector and to independent products of such sectors. Applying it inside a coupled graph would require coefficients that depend on the other loop momenta, which this implementation does not support.

Write $D_i = (\ell - q_i)^2 - m_i^2$ and let $S = \{i : a_i > 0\}$ contain the denominator positions. The q_i in this appendix are fixed external shifts; the infinitesimal prescription is understood. Choose $r \in S$, set $k = \ell - q_r$, and define $R_i = q_i - q_r$. The vectors R_i with $i \in S \setminus \{r\}$ span the external directions seen by the denominators. Their Gram matrix and their contractions with k are

$$G_{ij} = R_i \cdot R_j, \qquad u_i = k \cdot R_i = \frac{D_r - D_i + m_r^2 - m_i^2 + R_i^2}{2}. \tag{21}$$

The second equality follows by expanding $D_i = (k - R_i)^2 - m_i^2$ and subtracting it from $D_r = k^2 - m_r^2$. Assume that G is invertible, and split k into its projection onto this span and its perpendicular part:

$$k_\parallel^\mu = \sum_{i,j \in S\setminus\{r\}} R_i^\mu (G^{-1})_{ij} u_j, \qquad k_\perp = k - k_\parallel. \tag{22}$$

Contracting with each R_i gives $k_\perp \cdot R_i = 0$. Contracting the parallel part with itself gives $k_\parallel^2 = u^T G^{-1} u$, so

$$T(D) := k_\perp^2 = D_r + m_r^2 - u^T G^{-1} u. \tag{23}$$

Empty matrices and sums give zero when S has one element.

An inverse propagator D_j with $a_j < 0$ appears in the numerator and can be resolved into the same components. Define $v_{ji} = R_j \cdot R_i$, $R_{j,\perp} = R_j - \sum_{i,h} R_i (G^{-1})_{ih} v_{jh}$, and $z_j = k_\perp \cdot R_{j,\perp}$. Direct substitution gives

$$D_j = B_j(D) - 2z_j, \tag{24}$$

$$B_j(D) = D_r + m_r^2 + R_j^2 - m_j^2 - 2v_j^T G^{-1} u, \tag{25}$$

$$H_{jh} := R_{j,\perp} \cdot R_{h,\perp} = R_j \cdot R_h - v_j^T G^{-1} v_h. \tag{26}$$

Thus the remaining task is to average products of the z_j over perpendicular directions. All positive-power denominators depend on $k_\perp$ only through $T(D)$, so this angular average leaves the denominator unchanged.

F.2 Angular moments and the finite numerator expansion

Let $\nu = d - |S| + 1$ be the dimension perpendicular to the denominator span. Rotational symmetry makes odd angular moments vanish. For $2h$ perpendicular components, the invariant symmetric

tensor is the sum of all pairings of metric tensors, multiplied by a coefficient $C_h T^h$. Tracing two indices gives

$$C_h(\nu+2h-2)=C_{h-1},\qquad C_0=1,\qquad C_h=\frac{1}{\prod_{b=0}^{h-1}(\nu+2b)}. \tag{27}$$

The trace factor contains ν from the pairing of the traced indices and $2h-2$ from their pairings with the other indices. Contracting the tensor with the external perpendicular vectors therefore gives

$$\left\langle \prod_{t=1}^{2h} z_{j_t} \right\rangle = \frac{T(D)^h}{\prod_{b=0}^{h-1}(\nu+2b)} \sum_{\mathcal{P}} \prod_{(s,t)\in\mathcal{P}} H_{j_s j_t}. \tag{28}$$

Here $\mathcal{P}$ runs over the distinct pairings of the $2h$ positions, including their multiplicities when an index repeats. In particular,

$$\langle z_i z_j\rangle = \frac{T H_{ij}}{\nu},\qquad \langle z_i z_j z_k z_l\rangle = \frac{T^2(H_{ij}H_{kl}+H_{ik}H_{jl}+H_{il}H_{jk})}{\nu(\nu+2)}. \tag{29}$$

One can first derive these relations after a Wick rotation in an integer dimension, where the angular average is well defined, and then continue the rational tensor coefficients in d. This uses the dimensional-regularization convention of Eq. (1) [1]. Poles in d remain exact rational poles.

For $b_j=-a_j>0$, expand $\prod_{j:a_j<0}(B_j-2z_j)^{b_j}$ and insert Eq. (28). The result is a finite polynomial in the positive-sector denominators:

$$\left\langle \prod_{j:a_j<0} D_j^{b_j} \right\rangle = \sum_{\boldsymbol{t}} c_{\boldsymbol{t}}(d) \prod_{i\in S} D_i^{t_i},\qquad I_{\boldsymbol{a}} = \sum_{\boldsymbol{t}} c_{\boldsymbol{t}}(d) I_{\boldsymbol{a}_S-\boldsymbol{t}}. \tag{30}$$

The vector $\boldsymbol{a}_S$ equals $\boldsymbol{a}$ on S and vanishes elsewhere; the t_i are nonnegative and vanish outside S. The coefficients also depend rationally on the fixed kinematics. Each power $D_i^{t_i}$ cancels the corresponding denominator power. If a cancellation creates a new negative index, that line leaves the positive sector. Repeating the angular step therefore decreases the positive-sector size whenever a numerator remains. A sector with no denominators is scaleless and vanishes in dimensional regularization [1]. The process terminates in scalar sectors with nonnegative indices.

F.3 Lowering powers before removing the numerator

A second exact route starts with IBP. Define $A=\sum_j a_j$ and $Y_{ij}=m_i^2+m_j^2-(q_i-q_j)^2$. Expanding the total derivative with vector $\ell-q_i$ gives

$$0 = dI_{\boldsymbol{a}} - \sum_j a_j \int \frac{d^d\ell}{i\pi^{d/2}} \frac{2(\ell-q_i)\cdot(\ell-q_j)}{D_j} \prod_h D_h^{-a_h}, \tag{31}$$

$$2(\ell-q_i)\cdot(\ell-q_j) = D_i + D_j + Y_{ij}, \tag{32}$$

$$0 = (d-A)I_{\boldsymbol{a}} - \sum_j a_j I_{\boldsymbol{a}+\boldsymbol{e}_j-\boldsymbol{e}_i} - \sum_j a_j Y_{ij} I_{\boldsymbol{a}+\boldsymbol{e}_j}. \tag{33}$$

Here $\boldsymbol{e}_j$ is the unit index vector. This derivation holds for signed integer indices, so it includes numerator terms. Let $N_-=\{j:a_j<0\}$. Moving the negative-index terms to the right gives, for $i\in S$,

$$\sum_{j\in S} a_j Y_{ij} I_{\boldsymbol{a}+\boldsymbol{e}_j} = F_i(\boldsymbol{a}), \tag{34}$$

$$F_i(\boldsymbol{a}) = (d-A)I_{\boldsymbol{a}} - \sum_{j:a_j\neq 0} a_j I_{\boldsymbol{a}+\boldsymbol{e}_j-\boldsymbol{e}_i} - \sum_{j\in N_-} a_j Y_{ij} I_{\boldsymbol{a}+\boldsymbol{e}_j}. \tag{35}$$

For target $\boldsymbol{b}$ with $b_k > 1$, choose $\boldsymbol{a} = \boldsymbol{b} - \boldsymbol{e}_k$. If Y_S is invertible, multiply this system by its inverse to obtain

$$I_{\boldsymbol{b}} = \frac{1}{a_k} \sum_{i \in S} (Y_S^{-1})_{ki} F_i(\boldsymbol{a}). \tag{36}$$

Every integral on the right has a smaller sum of positive indices than $\boldsymbol{b}$, and the numerator rank does not increase. When the positive powers reach one, the angular step removes the remaining numerator. With no negative indices, Eq. (36) reduces to the scalar Cayley recurrence derived earlier. These identities give a finite reduction to unit-power scalar sectors. They produce a spanning set; special kinematics can introduce further relations between its elements.

F.4 Keeping the dimension poles in factored form

The preceding identities restrict the form of every dimension-dependent denominator. The inverse Gram and Cayley matrices have rational entries at the chosen kinematic point, and Eq. (36) introduces only polynomial dependence on d. Equation (28) introduces products of linear factors $d + j$, where j is an integer. By induction through the recurrence, each coefficient therefore has the form

$$r(d) = \frac{P(d)}{\prod_j (d+j)^{n_j}}, \qquad P(d) \in \mathbb{Q}[d], \quad n_j \in \mathbb{Z}_{\geq 0}. \tag{37}$$

The finite products contain only factors introduced by the angular moments. For addition, take Q to be the least common multiple of two such denominators Q_1 and Q_2. Multiplying each summand by its missing factors gives

$$\frac{P_1}{Q_1} + \frac{P_2}{Q_2} = \frac{P_1(Q/Q_1) + P_2(Q/Q_2)}{Q}. \tag{38}$$

The exponent of $d+j$ in Q is the larger of its two exponents. Multiplication adds these exponents. A factor $d+j$ cancels exactly when $P(-j) = 0$, because division with remainder gives $P(d) = (d+j)\widetilde{P}(d) + P(-j)$. We repeat this cancellation for each retained factor. These operations preserve the exact rational function and require only rational polynomial coefficients and integer factor multiplicities. They avoid general rational-function simplification in the new path. The earlier arithmetic checks compare these coefficients with a general rational-function implementation and native Kira reductions. The current application retains the fixed numerator-first recurrence; its new learned decision concerns native seed regions.

G Exact reduction at a zero two-point invariant

A singular Gram matrix prevents the projection used by the generic tensor recurrence. For a two-point sector with $p^2 = 0$, Feynman parameters give a direct reduction to tadpoles. Degenerate one-loop kinematics require such separate identities [7]. We derive the identity used in this study below. It applies to a polynomial numerator and two positive denominator powers; it does not assume that the two masses differ.

G.1 Scalar integrals and mass derivatives

Let x and y denote squared masses, and define

$$T(x) = \int \frac{d^d\ell}{i\pi^{d/2}} \frac{1}{\ell^2 - x + i0}, \qquad B_{ab}(x,y) = \int \frac{d^d\ell}{i\pi^{d/2}} \frac{1}{(\ell^2 - x + i0)^a [(\ell - p)^2 - y + i0]^b}\bigg|_{p^2=0}. \tag{39}$$

Here a, b are positive integers. Dimensional regularization sets $T(0) = 0$. With the normalization in Eq. (1), Wick rotation and the radial beta integral give [1]

$$T(x) = -\Gamma(1 - d/2)(x - i0)^{d/2-1}. \tag{40}$$

For $a = b = 1$, combine the denominators with a parameter z and shift $\ell = k + (1 - z)p$. The combined squared mass is $M^2(z) = zx + (1 - z)y$, because $p^2 = 0$. Integrating over k and then z gives

$$\begin{aligned} B_{11}(x, y) &= \Gamma(2 - d/2) \int_0^1 [zx + (1 - z)y - i0]^{d/2-2}\, dz \\ &= \frac{T(x) - T(y)}{x - y}, \qquad x \neq y. \end{aligned} \tag{41}$$

The second equality follows from the elementary antiderivative of a power and $\Gamma(2 - d/2) = (1 - d/2)\Gamma(1 - d/2)$. Differentiating a propagator with respect to its squared mass raises its power. Therefore

$$B_{ab}(x, y) = \frac{\partial_x^{a-1} \partial_y^{b-1} B_{11}(x, y)}{(a - 1)!(b - 1)!}. \tag{42}$$

This operation preserves a rational combination of $T(x)$ and $T(y)$ because $\partial_x T(x) = (d/2 - 1)T(x)/x$. We take the derivatives before imposing relations between the masses. If $x = y \neq 0$, the combined mass is independent of z, and the beta integral instead gives

$$B_{ab}(x, x) = T_{a+b}(x), \qquad T_n(x) = \frac{\prod_{j=1}^{n-1}(d/2 - j)}{(n - 1)!\, x^{n-1}} T(x). \tag{43}$$

The formula for T_n also follows by differentiating Eq. (40). These identities support independent symbolic squared masses as well as rational values. A relation involving distinct symbolic masses is a generic rational identity; equal masses use Eq. (43) directly.

G.2 Polynomial numerators

Let $P(\ell)$ be a polynomial numerator. Combining the two denominators gives

$$B_{ab}[P] = \frac{\Gamma(A)}{\Gamma(a)\Gamma(b)} \int_0^1 dz\, z^{a-1}(1 - z)^{b-1} \int \frac{d^d k}{i\pi^{d/2}} \frac{P(k + (1 - z)p)}{[k^2 - M^2(z) + i0]^A}, \quad A = a + b. \tag{44}$$

For an auxiliary inverse propagator $(\ell - q_j)^2 - m_j^2$, put $w_j = (1 - z)p - q_j$. Its numerator factor becomes $k^2 + 2k \cdot w_j + w_j^2 - m_j^2$. Both $w_i \cdot w_j$ and $w_j^2 - m_j^2$ are polynomials of degree at most one in z when $p^2 = 0$.

Rotational covariance gives zero for odd moments of k. Tracing two indices of an even moment determines its normalization recursively: if its pairing coefficient is C_h, then $C_h(d+2h-2) = C_{h-1}$ and $C_0 = 1$. Consequently,

$$\left\langle \prod_{j=1}^{2h} (k \cdot w_j) \right\rangle = \frac{(k^2)^h}{\prod_{r=0}^{h-1}(d + 2r)} \sum_{\mathcal{P}} \prod_{(i,j)\in\mathcal{P}} w_i \cdot w_j, \tag{45}$$

where $\mathcal{P}$ runs over pairings of the $2h$ positions. The trace recurrence follows because one pairing contributes d and the remaining contractions contribute $2h - 2$. This also derives the full-dimensional form of the angular identity used in Appendix F.

After the finite numerator expansion and angular average, define $c_{ht}(d)$ by

$$\langle P(k + (1 - z)p) \rangle = \sum_{h,t} c_{ht}(d)(k^2)^h z^t. \tag{46}$$

These coefficients also depend on the fixed external scalar products and numerator masses. They contain no loop momentum. To integrate each term, Wick rotate the radial integral, set $u = k_E^2/M^2$, and use $\int_0^\infty u^{\alpha-1}(1+u)^{-A}du = \Gamma(\alpha)\Gamma(A-\alpha)/\Gamma(A)$ with $\alpha = d/2+h$. This gives

$$J_{Ah}(M^2) = \int \frac{d^dk}{i\pi^{d/2}} \frac{(k^2)^h}{(k^2-M^2+i0)^A} = (-1)^{A-h}\frac{\Gamma(d/2+h)\Gamma(A-h-d/2)}{\Gamma(d/2)\Gamma(A)}(M^2-i0)^{d/2+h-A}. \tag{47}$$

The initial convergence region defines the integrals; analytic continuation gives the dimensional-regularization identity outside that region [1].

For $x \neq y$, set $\Delta = x-y$, $\rho = h-A+1$, and $\lambda = d/2-1$. Expand $(1-z)^{b-1}$ and then substitute $u = y+\Delta z$. For an integer $k \geq 0$,

$$\int_0^1 z^k(y+\Delta z)^{\lambda+\rho}\,dz = \frac{1}{\Delta^{k+1}}\sum_{j=0}^{k}\binom{k}{j}(-y)^{k-j}\frac{x^{\lambda+\rho+j+1}-y^{\lambda+\rho+j+1}}{\lambda+\rho+j+1}. \tag{48}$$

This follows by expanding $(u-y)^k$ and integrating each power of u. Dividing by the tadpole normalization in Eq. (40), define

$$R_h = \frac{(-1)^{A-h+1}(d/2)_h(1-d/2)_{A-h-1}}{(a-1)!(b-1)!}, \qquad (z)_n = \frac{\Gamma(z+n)}{\Gamma(z)}. \tag{49}$$

For a negative integer n, the last ratio means the inverse finite product $1/\prod_{r=1}^{-n}(z-r)$. Equations (44)–(48) now give the explicit rational reduction

$$\begin{aligned} B_{ab}[P] = \sum_{h,t} c_{ht}(d)R_h \sum_{v=0}^{b-1}(-1)^v\binom{b-1}{v} \\ \times \sum_{j=0}^{k}\binom{k}{j}\frac{(-y)^{k-j}}{\Delta^{k+1}}\frac{x^{\rho+j+1}T(x)-y^{\rho+j+1}T(y)}{\lambda+\rho+j+1}, \\ k = a-1+t+v. \end{aligned} \tag{50}$$

If one mass is zero, its endpoint vanishes in a convergence region and hence by analytic continuation; the implementation omits that scaleless endpoint before evaluating powers. If both masses vanish, the sector is scaleless. For equal nonzero masses, use the beta integral directly:

$$B_{ab}[P] = \sum_{h,t} c_{ht}(d)R_h\,\frac{\Gamma(a+t)\Gamma(b)}{\Gamma(A+t)}x^{h-A+1}T(x). \tag{51}$$

Thus the equal-mass rule has no division by $x-y$. A sector with one positive denominator uses Eq. (47) and the same angular expansion without a Feynman parameter.

At rational kinematics, the coefficients have polynomial numerators in d and products of integer-shift linear factors $d+j$ in their denominators. The finite products in Eqs. (45) and (49), and the linear denominator in Eq. (48), prove this statement. We therefore use the exact factored arithmetic of Appendix F. The implemented tensor domain has at most five coordinates, numerator rank at most four, and at most four extra positive powers. Symbolic masses currently use the scalar derivative identity. Inputs outside these conditions continue through the existing exact rules and Kira.

H Connected one-scale tree reductions

We derive the previously retained tree rules to state precisely which high-loop structures they cover. Let $\mathcal{T}$ be a rooted tree with loop momenta $\ell_0, \ldots, \ell_{L-1}$ and parent $\pi(j)$ for every nonroot

vertex. A massless two-point family is

$$J_{\mathcal{T}} = \int \prod_{j=0}^{L-1} \frac{d^d \ell_j}{i\pi^{d/2}} \frac{1}{(\ell_0^2)^{a_0}[(\ell_0 - p)^2]^c} \prod_{j=1}^{L-1} \frac{1}{(\ell_j^2)^{a_j}[(\ell_j - \ell_{\pi(j)})^2]^{v_j}}, \qquad p^2 = s \neq 0. \tag{52}$$

The massive vacuum version replaces the two root factors by $(\ell_0^2 - m^2)^{-c}$ and has no external momentum. Each edge power v_j, root power c, and leaf self power a_j is positive. Internal self powers can be nonpositive. Auxiliary powers vanish. The implementation admits $L \leq 13$ and absolute physical powers at most sixteen. The derivation itself applies to finite trees with meromorphically defined integrals.

Temporarily use negative massless denominators. Feynman parametrization of a massless bubble, translation of its loop momentum, and a radial beta integral give

$$\begin{aligned} G(u, v) &= \frac{\Gamma(u + v - d/2)}{\Gamma(u)\Gamma(v)} \int_0^1 z^{d/2-v-1}(1 - z)^{d/2-u-1}\, dz \\ &= \frac{\Gamma(u + v - d/2)\Gamma(d/2 - u)\Gamma(d/2 - v)}{\Gamma(u)\Gamma(v)\Gamma(d - u - v)}. \end{aligned} \tag{53}$$

The remaining scale factor is $(-P^2 - i0)^{d/2-u-v}$ for bubble momentum P. The beta identity used here follows by the same radial and parameter integrations as Appendix G; analytic continuation supplies the dimensionally regularized relation [1]. Recursive subintegral integration has established applications to massless propagator integrals [14].

For each nonroot vertex, define

$$u_j = a_j + \sum_{h:\pi(h)=j} \sigma_h, \qquad \sigma_j = u_j + v_j - d/2. \tag{54}$$

Integrating a leaf gives $G(u_j, v_j)$ and transfers power σ_j to its parent. Repeating from the leaves to the root proves the product rule by induction. In the two-point case, set $\alpha = a_0 + \sum_{h:\pi(h)=0} \sigma_h$; its dimensionless coefficient is

$$C_{\mathcal{T}} = G(\alpha, c) \prod_{j=1}^{L-1} G(u_j, v_j). \tag{55}$$

The total physical power $A = \sum_j a_j + \sum_{j>0} v_j + c$ gives scale power $(-s - i0)^{Ld/2-A}$. Let $M_{\mathcal{T}}$ set all physical powers to one and let C_0 be its Gamma product. Restoring the original propagator signs and dividing by this master gives

$$J_{\mathcal{T}} = s^{-(A-2L)} \frac{C_{\mathcal{T}}}{C_0} M_{\mathcal{T}}. \tag{56}$$

The sign $(-1)^{A-2L}$ from the propagator convention cancels that from $(-s)^{-(A-2L)}$.

For a massive vacuum root, put $\sigma_0 = \sum_{h:\pi(h)=0} \sigma_h$. Wick rotation and the substitution $x = m^2 t$ give

$$\frac{1}{\Gamma(d/2)} \int_0^\infty \frac{x^{d/2-\sigma_0-1}}{(x + m^2)^c}\, dx = (m^2)^{d/2-\sigma_0-c} \frac{\Gamma(d/2 - \sigma_0)\Gamma(\sigma_0 + c - d/2)}{\Gamma(d/2)\Gamma(c)}. \tag{57}$$

Multiply the dimensionless Gamma ratio by the nonroot bubble factors to define $C_{\mathcal{T}}^V$. There are $P = 2L - 1$ physical lines and $A = c + \sum_{j>0}(a_j + v_j)$. Dividing by the unit-power vacuum master $M_{\mathcal{T}}^V$ yields

$$V_{\mathcal{T}} = (-m^2)^{-(A-P)} \frac{C_{\mathcal{T}}^V}{C_0^V} M_{\mathcal{T}}^V. \tag{58}$$

In either family, a subtree with n_j vertices has $u_j = U_j - (n_j - 1)d/2$ with integer U_j. This follows from Eq. (54) by induction. Target and master Gamma arguments therefore have equal slopes in

d and integer differences in their constant terms. Repeated use of $\Gamma(z+1) = z\Gamma(z)$ gives

$$\frac{\Gamma(z+n)}{\Gamma(z)} = \begin{cases} \prod_{k=0}^{n-1}(z+k), & n > 0, \\ 1, & n = 0, \\ [\prod_{k=1}^{-n}(z-k)]^{-1}, & n < 0. \end{cases} \tag{59}$$

Thus the target-to-master coefficients are rational in d and the one scale. Cancelling equal Gamma factors before expanding these finite products preserves the identity and avoids repeated symbolic work. This proves the reductions for the stated trees; additional coupled propagators or auxiliary numerators require another reduction.

I Direct evaluation of vacuum factors

We first identify a case that requires no master integral. If all masses vanish, there is no external momentum, and every active inverse propagator is the square of a homogeneous linear combination of loop momenta, the integrand has no dimensional scale. After Wick rotation, let R be the common radial coordinate for all loop momenta. The measure and denominators give radial power $R^{\omega-1}$ with $\omega = Ld - 2\sum_j a_j$. Split the radial integral at an arbitrary $\mu > 0$ and continue its two pieces separately:

$$\int_0^\mu R^{\omega-1}\, dR + \int_\mu^\infty R^{\omega-1}\, dR = \frac{\mu^\omega}{\omega} - \frac{\mu^\omega}{\omega} = 0. \tag{60}$$

The two pieces initially converge on opposite half-planes and their meromorphic continuations cancel. Angular singularities can be treated by analytic regularization before the same scaling argument. This is the standard scaleless-integral prescription in dimensional regularization [1]. It gives the combined integral as zero; separating ultraviolet and infrared poles requires retaining the two regulated contributions separately. The implementation checks the masses and momentum homogeneity, rejects a zero denominator with positive power, and accepts up to thirteen loops and 128 coordinates. It excludes cut denominators and symbolic-index inputs. Couplings between loop momenta are permitted for this zero-integral rule.

Consider a loop momentum k that occurs only in denominators $k^2 - x_j + i0$, where x_j is a squared mass. There is no external momentum in this component. Denominator powers can be positive, zero, or negative. Equal masses first combine by addition of their powers. With $z = k^2$, let A_x be the resulting integer power at each distinct squared mass x. The integrand is the rational function

$$F(z) = \prod_x (z-x)^{-A_x}. \tag{61}$$

For a mass with $A_x > 0$, multiply by $(z-x)^{A_x}$ and expand the remaining analytic factor at $z = x$. Its first A_x Taylor coefficients give the principal part at that pole. Subtracting every principal part leaves a rational function without a finite pole, which is a polynomial $P(z)$. Thus

$$F(z) = P(z) + \sum_{x:A_x>0} \sum_{r=1}^{A_x} \frac{c_{xr}}{(z-x)^r}, \tag{62}$$

$$c_{xr} = \frac{1}{(A_x-r)!} \frac{d^{A_x-r}}{dz^{A_x-r}} \prod_{y\neq x} (z-y)^{-A_y} \bigg|_{z=x}. \tag{63}$$

This derivation also covers negative powers, since they supply polynomial factors in the product. For distinct rational masses, the coefficients are rational numbers. The implementation obtains

them by multiplying truncated Taylor series, so it avoids general symbolic differentiation. A factor with exponent $-A_y$ has coefficient

$$[t^n](x-y+t)^{-A_y} = \binom{-A_y}{n}(x-y)^{-A_y-n}. \tag{64}$$

The binomial series proves Eq. (64); only $0 \le n < A_x$ is needed in Eq. (63). Equal masses must be combined before this expansion, which avoids division by a vanishing mass difference.

Polynomial terms in k and massless vacuum integrals have no scale and vanish in dimensional regularization [1]. For a nonzero squared mass x, define the normalized tadpole

$$T_r(x) = \int \frac{d^dk}{i\pi^{d/2}}(k^2 - x + i0)^{-r}. \tag{65}$$

To evaluate it, first take $x > 0$ and a convergent Euclidean domain. The Schwinger representation follows from the defining Gamma integral after the change of variable $u = \alpha(q^2 + x)$:

$$(q^2+x)^{-r} = \frac{1}{\Gamma(r)}\int_0^\infty \alpha^{r-1}e^{-\alpha(q^2+x)}\,d\alpha. \tag{66}$$

The Gaussian momentum integral is $\int d^dq\, e^{-\alpha q^2} = \pi^{d/2}\alpha^{-d/2}$. Inserting this in Eq. (66) and setting $u = \alpha x$ gives $x^{d/2-r}\Gamma(r-d/2)/\Gamma(r)$. Wick rotation supplies $(-1)^r$ for the denominator convention in Eq. (65). Analytic continuation therefore gives

$$T_r(x) = (-1)^r\frac{\Gamma(r-d/2)}{\Gamma(r)}(x-i0)^{d/2-r}, \qquad \frac{T_r(x)}{T_1(x)} = \frac{\prod_{j=1}^{r-1}(d/2-j)}{(r-1)!x^{r-1}}. \tag{67}$$

The second equality follows by applying $\Gamma(z+1) = z\Gamma(z)$ repeatedly to the first. The empty product at $r = 1$ is one. Applying Eq. (67) term by term in Eq. (62) proves both the reduction to unit tadpoles and their full analytic evaluation. The result is a meromorphic expression in d; ultraviolet or infrared poles are retained. The same tadpole value also evaluates any unit tadpole returned by the retained zero-invariant tensor formulas. A remaining master with several denominators still requires its own evaluation.

Independent components multiply, as proved in Appendix A. The direct implementation accepts one through thirteen such loop momenta, at most eight denominators per component, individual absolute powers at most eight, and total positive power after equal-mass combination at most 48 per component. Masses are nonnegative rational numbers. Denominators with nonzero target power must carry one of those individual momenta with coefficient $+1$ or -1. These tests exclude couplings between components and specify the actual scope of the rule.

The connected one-mass vacuum trees have a separate analytic evaluation. Appendix H derives their dimensionless Gamma product $C^V_{\mathcal{T}}$. With total physical power A, Wick rotation of all integer-power denominators gives

$$V_{\mathcal{T}} = (-1)^A(m^2 - i0)^{Ld/2-A}C^V_{\mathcal{T}}. \tag{68}$$

The exponent follows either from the final radial integral or from the mass dimension of the L measures and A denominators. For a unit-power master, $A = P = 2L-1$. The application now returns its explicit Gamma arguments and scale exponent together with the rational reduction. This exposes the analytic master value already implied by the tree derivation and avoids a separate master evaluation for these families.

J Inputs and checks for the direct-evaluation comparison

The two-mass vacuum grid contains one request at each $L = 1, \ldots, 13$. At loop order L, each independent component has squared masses $(h+2, h+7)$ with $h = 23011+L$. Its four final target

power pairs are $(2,1),(3,1),(2,3),(3,-1)$, and each full target repeats the same pair across all L components. These are products of one-loop factors. The connected vacuum grid has 34 requests: chains at $L=1,\ldots,13$, stars at $L=3,\ldots,13$, and binary trees at $L=4,\ldots,13$. Their parent maps are $\pi(j)=j-1$, $\pi(j)=0$, and $\pi(j)=\lfloor (j-1)/2\rfloor$, respectively. These starting orders omit coincident small-loop routings. The root squared mass remains symbolic. Each final request has root power two, root power three, root and final edge powers two, and root power four, with every other physical power one. The one-loop request instead uses powers two through five. The unit-power integral remains the master. The profiles are $23111+L$, $23211+L$, and $23311+L$ for the three shapes and fix any auxiliary construction choices. These requests test preservation of the connected identity and the earlier symbolic single-loop dispatch.

The repeated-mass coverage grid uses four denominators per component with squared masses $(h+2,h+7,h+2,0)$, where $h=23511+L$. The four power assignments are $(1,1,2,0)$, $(2,1,1,1)$, $(1,2,0,-2)$, and $(0,0,0,2)$. This tests repeated masses, a massless pole, a polynomial numerator, and a scaleless target. It is checked through all thirteen loop orders, with no speed claim against an unfinished or unsupported native calculation.

The scaleless coverage grid has one family for each $L=1,\ldots,13$, with inverse propagators ℓ_i^2 and $(\ell_i-\ell_j)^2$ for all $i<j$. It therefore contains $L(L+1)/2$ coordinates and coupled loop momenta for $L>1$. The three targets set all powers to one, raise the first power to two, or change the first power to minus one. Equation (60) supplies the reference zero. Native one- and two-loop checks test the same rule separately.

The three retained zero-invariant checks use the unequal, equal, and mixed mass choices of Eq. (16), with profiles $23211, 23212, 23213$, numerator rank two, and two excess positive powers. The second excess power is placed on the first physical denominator. The three ordinary tensor checks use Eq. (17) with $h=23311+P$ for $P=2,3,4$. Their numerator rank and excess positive power are both two, with one of the excess powers placed on the first physical denominator.

The native inputs use Eqs. (18) and (19). Development contains two batches per graph at profile 22307; training contains twelve per graph at 22343; validation contains six per graph at 22349; and the preceding candidate test contains eight per graph at 22367. The final deployment test uses eight fresh whole batches per graph at profile 24443, excluding all batches in those earlier sets. Each batch has two to six distinct targets of the form stated in Appendix D. Whole batches are disjoint across these sets and exclude the preceding native collections. The final native model is retained without refitting; the fresh batches therefore enter only the final test. Individual target vectors can recur, and both graph types occur in training and testing.

An independent symbolic partial-fraction calculation checks the vacuum grid. Random rational-function checks verify that subtracting the proposed poles leaves only a polynomial. Convergent Euclidean radial quadrature checks the tadpole normalization, and a native IBP comparison checks its raised-power recurrence. Native box and ordinary tensor coefficients are checked against independently prepared Kira reductions. The connected tree comparison checks preservation against the preceding exact rule; its mathematical justification is the separate derivation in Appendix H.

References

[1] Stefan Weinzierl. *Feynman Integrals: A Comprehensive Treatment for Students and Researchers*. UNITEXT for Physics. Springer, 2022. arXiv:2201.03593; publisher DOI.

[2] K. G. Chetyrkin and F. V. Tkachov. Integration by parts: The algorithm to calculate β-functions in 4 loops. *Nuclear Physics B*, 192:159–204, 1981. publisher DOI.

[3] Philipp Maierhöfer, Johann Usovitsch, and Peter Uwer. Kira—a Feynman integral reduction program. *Computer Physics Communications*, 230:99–112, 2018. arXiv:1705.05610.

[4] Fabian Lange, Johann Usovitsch, and Zihao Wu. Kira 3: Integral reduction with efficient seeding and optimized equation selection. *Computer Physics Communications*, 322:109999, 2026. arXiv:2505.20197.

[5] Zhuo-Yang Song, Tong-Zhi Yang, Qing-Hong Cao, Ming-xing Luo, and Hua Xing Zhu. Explainable AI-assisted optimization for Feynman integral reduction, 2025. arXiv:2502.09544.

[6] Justin Berman, Francois Charton, Andres Luna, Matthias Wilhelm, and Mao Zeng. Efficient ai-inspired reduction of feynman integrals via tube seeding, 2026. arXiv:2606.10698.

[7] A. Denner and S. Dittmaier. Reduction schemes for one-loop tensor integrals. *Nuclear Physics B*, 734:62–115, 2006.

[8] Chang Hu, Bo Feng, Jiyuan Shen, and Yaobo Zhang. General one-loop generating function by integration-by-part relations. *Physical Review D*, 111:076026, 2025.

[9] Stefano Laporta. High-precision calculation of multiloop Feynman integrals by difference equations. *International Journal of Modern Physics A*, 15:5087–5159, 2000. arXiv:hep-ph/0102033.

[10] Jonas Klappert, Fabian Lange, Philipp Maierhöfer, and Johann Usovitsch. Integral reduction with Kira 2.0 and finite field methods. *Computer Physics Communications*, 266:108024, 2021. arXiv:2008.06494.

[11] Leo Breiman, Jerome H. Friedman, Richard A. Olshen, and Charles J. Stone. *Classification and Regression Trees*. Chapman & Hall, 1984. Publisher record.

[12] Leo Breiman. Random forests. *Machine Learning*, 45:5–32, 2001.

[13] Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. publisher DOI.

[14] Ben Ruijl, Takahiro Ueda, and Jos A. M. Vermaseren. Forcer, a FORM program for the parametric reduction of four-loop massless propagator diagrams, 2017.